1.7 Stress Strain Curve For Mild Steel: Curve (Fig. 1.7) for a round mild steel bar of which ordinate represents the stress intensities and the abscissa the corresponding strains. The points worth noting are:

- The line OA being straight. OA is called the proportional limit upto which Hooke's law is applicable i.e. stress is proportional to strain up to A.
- Mild steel remains elastic a little beyond A i.e. upto point A, so the proportional limit and elastic limit are not same for mild srteel but same for rest of the materials.
- Upon loading beyond this point the elongation increases more rapidly and the diagram becomes curved. At B a sudden elongation of the bar takes place without an appreciable increase in the tensile force. The phenomenon is called ‘Yielding’ of the metal. The stress corresponding to this ‘B’ point is called yield point stress.

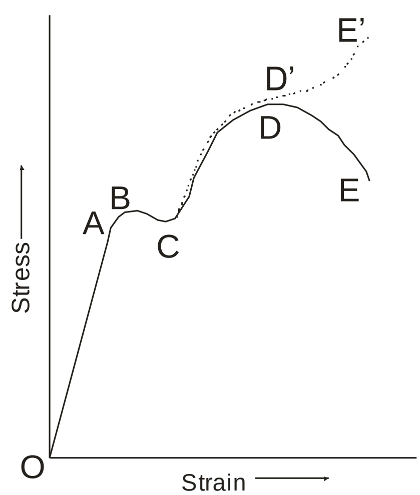

Fig. 1.7

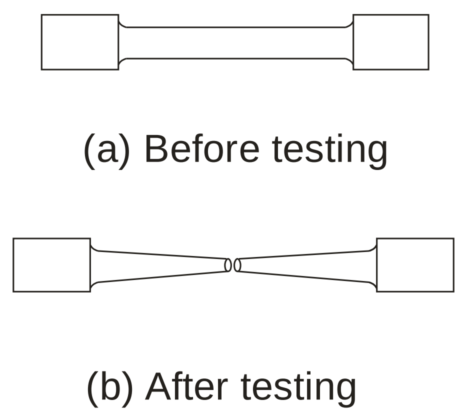

Fig. 1.8

- Upon further stretching of the bar, the material recovers its resistance and the tensile force increases with the elongation upto point ‘D’, where the force attains its maximum value and the process of necking begins.

  The corresponding stress is called the ‘Ultimate Stress’. Necking is displayed by contraction of the specimen. Fig. 1.8.
- Beyond the point D elongation of the bar takes place with a diminution of the load and fracture finally occurs at a load corresponding to point E of diagram. The corresponding stress is called the ‘Breaking stress’.

The stress-strain curve is drawn experimentally and it is unique for every metal. Hence the points observed on the curve are the properties of the metal and unique for every metal. So the point A corresponds to **limit of elasticity**, point B corresponds to **Yield strength** and point D correspond to **Ultimate strength** of the metal.

Stretching of this bar is accompanied by lateral contraction but it is established for calculating the yield & ultimate point stress, hence initial cross-sectional area is used. If the load is divided by the original area of cross section it results in nominal stress which is less at breaking load than at the maximum load as indicated by E&D respectively. If actual area of cross section is used to find stress intensity, rupture or breaking stress is greater than at the maximum load. The dashed portion of the curve shows relation between actual stress intensity and strain. This is called true stress strain curve.

Fig. 1.9 represents a tensile test diagram for cast iron. The material has a very low proportional limit and has no definite yield point. For brittle materials stress-strain curve is as in Fig. 1.9. There is no yield point in case of brittle material.

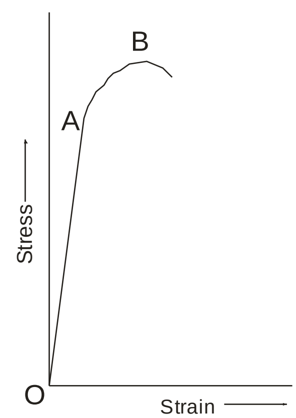

Fig. 1.9

<u>1.8 Factor Of Safety And Working Stresses</u>:

Working stress: The greatest calculated stress to which a part of machine or structure is ever subjected is called the working stress or allowable stress or permissible stress or design stress and the ratio

$$\frac{\text{Ultimate Strength}}{\text{Working Stress}} \text{ or } \frac{\text{Yield Stress}}{\text{Working Stress}} \quad \text{…..(1.11)}$$

is called Factor of Safety. The ultimate or yield strength is the property of the metal and it is known while selecting the metal for design purpose. For design calculations the factor of safety is to be assumed and design stress is to be calculated. Hence from the equation (1.11) working stress is the ratio

$$\frac{\text{Ultimate Strength}}{\text{Factor of safety}} \text{ or } \frac{\text{Yield Stress}}{\text{Factor of safety}} \quad \text{…..(1.12)}$$

Factor of safety depends upon:

1. **Nature of loading**: The factor of safety varies very greatly according to the nature of the stresses, whether constant, variable or alternating simple or compound.
2. **Homogeneity of materials used**: The factor of safety will be higher if the quality of material used is not of uniform properties.
3. **Degree of safety required:** If safety required is more than factor of safety will be higher. For example the areas where human hazards are involved we would require more safety like boiler applications.
4. **Degree of economy desired:** If we go for higher factor of safety the cost of the component will be more. Hence economical constraints are equally important while considering factor of safety.
5. **The accuracy of problem formulations:** If the estimation of stresses and load is not accurate we have to take higher factor of safety.
6. **Manufacturing processes:** If the quality of manufacturing methods used is good factor of safety required is lesser.

1.9 Elongation of Bars of Varying Cross Section

Fig 1.10 Shows a bar consisting of three lengths l_1 , l_2 , l_3 with cross section areas A_1, A_2, and A_3, Subjected to an axial load P. Though the force on each section is same stress induced in each section is different.

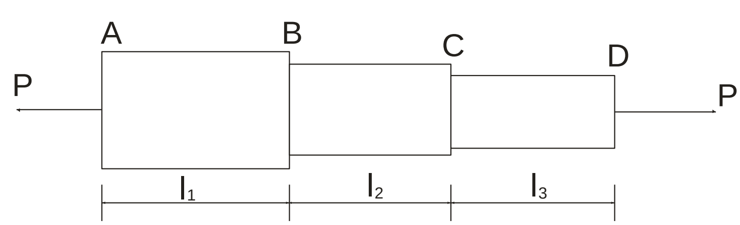

Fig. 1.10

Intensity of stress for section AB = σ_1 = P/A_1

Similarly intensity of stress for section BC = $\sigma_2 = P/A_2$

and intensity of stress for section CD = $\sigma_3 = P/A_3$

$\therefore$ Strain of section AB = $\in_1 = \dfrac{\sigma_1}{E}$

Where E – Modulus of elasticity

$\therefore$ Strain of section BC = $\in_2 = \dfrac{\sigma_2}{E}$

$\therefore$ Strain of section CD = $\in_3 = \dfrac{\sigma_3}{E}$

Change in length of section AB = $\delta l_1 = \in_1 l_1 = \dfrac{Pl_1}{A_1 E}$

Change in length of section BC = $\delta l_2 = \in_2 l_2 = \dfrac{Pl_2}{A_2 E}$

Change in length of section CD = $\delta l_3 = \in_3 l_3 = \dfrac{Pl_3}{A_3 E}$

$\therefore$ Total change in length of bar = $\delta l_1 + \delta l_2 + \delta l_3$

$$= \frac{P}{E}\left[\frac{l_1}{A_1} + \frac{l_2}{A_2} + \frac{l_3}{A_3}\right] \qquad \text{.....(1.13)}$$

1.12 **Principal Of Superposition**

Concept of free body diagram

If an elastic body is subjected to a no. of direct stresses at different sections along the length of the body, the deformation of individual sections can be very easily found if free body diagrams are drawn for the individual sections. According to principle of superposition the total deformation of the body will then be equal to the algebraic sum of the deformations of the individual sections. For example let us consider a bar subjected to different forces at different sections as in Fig. 1.14 (a). To draw the free body diagrams for AC all the forces to the left of the normal sections A to be considered. The net force is 20N, tensile considering the net force to right of C is 18 + 6 - 4 = 20N.

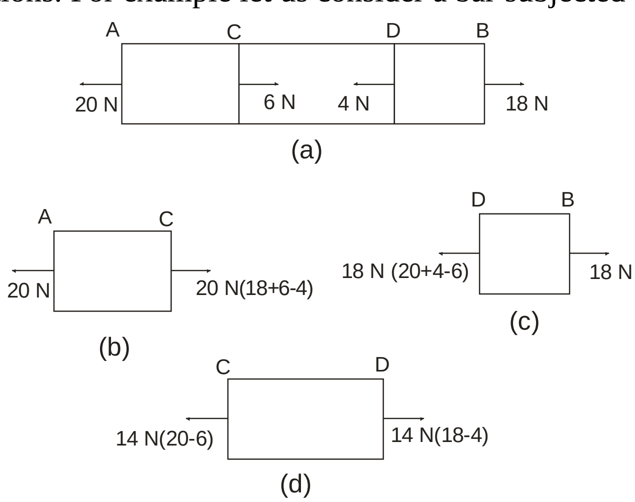

Fig. 1.14

Thus the section AC of length l_1, is subjected to tensile force of 20 N, as in Fig. 1.14 (b). Similarly, for the section CD of length l_2 consider the algebraic sum of the forces to the left of normal section at C or the sum of the forces to the right of the normal section at D. Net force on CD = 20 – 6 = 14 or 18 – 4 = 14. (Fig. 1.14 (d))

Similarly section DB is subjected to 18 N (Fig. 1.14 (c))

The extension of AC $\Delta_1 = \frac{P\,l_1}{AE} = \frac{20\,l_1}{A_1 E}$

CD $\Delta_2 = \frac{14\,l_2}{A_2 E}$

DB $\Delta_3 = \frac{18\,l_3}{A_3 E}$

Total extension of the bar AB will be given by

$$\Delta = \Delta_1 + \Delta_2 + \Delta_3 \qquad \text{…..(1.17)}$$